

User Story Document

Solar-R-Us LLC App and Product Backlog

Purpose

This document defines app-related user stories for the SRU solar water heating system. Each story includes role, need, value, acceptance criteria, and priority in the product backlog.

User Story #	Priority	Requirement - Role, Need, Value	Acceptance Criteria
001	1	As a commercial facility manager, I need to monitor water temperature, storage level, flow, and energy usage from the SRU app so I can control operating cost and ensure hot-water availability.	Given a configured commercial system, when the manager opens the dashboard, then current temperature, storage, flow, and energy-use status are displayed. Given readings change, when the system refreshes, then values update within the configured interval. Given a reading falls outside threshold, when the alert condition occurs, then the app displays a clear warning and recommended action.
002	2	As a homeowner, I need low-energy-use options and simple temperature controls so I can reduce monthly utility bills without needing technical knowledge.	Given the homeowner opens settings, when low-energy mode is selected, then the system applies energy-saving defaults. Given the homeowner changes target temperature, when the setting is saved, then the app confirms the new setting. Given a setting is unsafe or outside allowable limits, when the user submits it, then the app prevents the change and explains why.
003	3	As an installer, I need guided setup for evacuated tube, flat plate, solar-only, conventional heater, and instant heater configurations so I can complete installation accurately and reduce rework.	Given the installer starts setup, when a system type is selected, then the app displays configuration-specific steps. Given installation is complete, when validation is run, then the app confirms sensor, flow,

			storage, and temperature readings. Given the large commercial configuration is selected, when the app validates the model, then it recognizes the configuration and enables all supported features.
004	4	As a sales representative, I need an ROI and total cost of ownership calculator so I can explain financial value to commercial buyers in non-technical terms.	Given buyer inputs for utility cost, water usage, system type, and installation cost, when the calculator runs, then estimated savings, payback, and ROI are displayed. Given a comparison model is selected, when results are generated, then the SRU system is compared against conventional and competitor alternatives. Given assumptions are used, when results are shown, then assumptions are visible and exportable.
005	5	As a customer support agent, I need troubleshooting steps and system diagnostics so I can resolve customer issues quickly after launch.	Given a customer reports an issue, when the support agent enters symptoms or diagnostic codes, then recommended troubleshooting steps are shown. Given an issue is unresolved, when escalation is needed, then the app captures logs and customer/system details. Given repeated issues occur, when reports are reviewed, then product and support teams can identify trends.

Prioritized Product Backlog Summary

Rank	Backlog Item	Reason for Priority
1	Commercial monitoring dashboard	Core value for commercial launch and needed to validate operational savings.
2	Low-energy controls	Directly supports the value proposition of lower utility cost and easy use.
3	Guided installer setup	Reduces installation errors and

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		addresses setup problems found during beta testing.
4	ROI and total cost of ownership calculator	Supports sales enablement, marketing collateral, and buyer justification.
5	Support diagnostics and troubleshooting workflow	Improves post-launch readiness and customer support quality.

References Used

Course scenario and User Story/Product Backlog Worksheet provided by the Product Management Capstone assignment.